Euclidean → Hyperbolic Retraining Protocol

Status: DRAFT for Stewart's nod, 2026-10-27. Base: Muse Glimmer 30B, MLX Q4, ~17 GB. Submission deadline: Nov 4, 9:00 PM ET.

Bottom line

Retraining counts only if the delta is learned on a hyperbolic manifold, passes a deterministic geometry test, preserves general knowledge within 2%, and beats the Euclidean arms by 10x on the targeted reuse metric.

A Euclidean LoRA with hyperbolic labels does not count. A hyperbolic retrieval layer over frozen Euclidean weights does not count. API tokens do not count.

1. Definitions

- W = base Euclidean weights, Glimmer Q4.
- Δ_H = hyperbolic-native delta, parameterized by the hyperbolic graph.
- W_eff = W + Δ_H. Merged vs. adapter-only is bookkeeping, not identity.
- H = hyperbolic space, stored as Poincaré ball points, equivalently Lorentz vectors [x0 = sqrt(1 + |s|^2), s].
- d_H = hyperbolic distance. In the Lorentz model, d_H = arcosh(-<x, y>_L).
- Radius = generality. Center = shared knowledge. Boundary = private detail.
- P_gen = general-knowledge parameters in W. They are not overwritten; Δ_H is trained separately.
- P_novel = new parameters in Δ_H and in the map atoms. They are versioned separately from W.

2. Method: breaking Euclidean muscle memory

Euclidean muscle memory is the set of habits the base model learned in flat space:

- Cosine / dot-product similarity as relevance.
- Flat attention over flat vectors.
- Exact-prefix reuse or nothing.
- Euclidean SGD on a free low-rank delta.

Breaking it means replacing each habit, not relabeling it:

Habit	Euclidean	Hyperbolic-native
Similarity	Cosine / dot product	d_H, geodesic distance
Recall	Exact-prefix match	Geodesic traversal, graded near-miss reuse
Placement	Flat vector index	Radius = depth / generality
Forgetting	Score-based eviction	Radial drift, anchors hold, sweeper reclaims past dead radius
Update	Euclidean SGD on free Δ	Riemannian gradient, retraction, parallel transport on H

Preservation rule:

- W stays the source of general knowledge.
- Δ_H carries only the hyperbolic adaptation.
- Novel training goes into Δ_H and map atoms, never by overwriting W in place without a receipt.
- Every delta carries a receipt: base id, source tensors, batch order, seed, domain, gate scores.

Training shape, corrected:

- Batches are data batches, not per-shard training.
- A safetensors shard is an arbitrary file cut. It can land mid-layer.
- A training step needs the full forward / backward pass through the stack.
- Training fragments against a frozen rest gives wrong gradients.
- Per-shard is real only at fuse time: dequantize, add delta, requantize, shard by shard.
- First round is effective only if it is a full-stack batch pass, not a shard pass.

Steps:

1. Pull MLX Q4 Glimmer, verify index and tensor count (1,568 tensors on record).
2. Extract real representations: hidden states / embeddings from real Glimmer weights on real tasks. Log source tensor, task, date.
3. Place representations in H. Check Lorentz consistency and unit-ball bounds.
4. Train Δ_H in data batches, full forward, Riemannian updates, checkpointed.
5. Fuse shard by shard only when a gate is earned.
6. Run A / B / C ablations on identical data, identical order, greedy decoding, fixed seed.
7. Log every run. No gate, no claim.

3. Qualification gates

All gates must pass. A failed gate withdraws the claim it guards.

Gate	Name	Pass condition
G0	Provenance	Base id, weight source, tensor count, and delta receipt logged. No API-token-only provenance.
G1	Geometry acceptance	See Section 4. Deterministic, repeatable.
G2	Knowledge preservation	See Section 5. Per domain, and aggregated.
G3	Targeted gain	See Section 6. C only.
G4	Reproducibility	Same seed, same data, same batch order, greedy decoding. Gate numbers repeat within stated tolerance. No sampling in gate runs.

Gate order is fixed: G0, then G1, then G2, then G3. Do not skip to G3 if G1 or G2 fails.

4. Geometry acceptance benchmark (deterministic)

Purpose: verify the geometry actually changed, not just the labels.

Input, fixed and frozen before any run:

- A fixed set of triples from the map tree: parent, child, sibling, unrelated.
- Ancestry is known from the tree, not inferred from the model.
- Depth is known from radius: parent radius < child radius.

Tests, all deterministic, greedy, fixed seed:

1. Ordering: d_H(parent, child) < d_H(parent, unrelated).
2. Sibling ordering: d_H(parent, child) < d_H(parent, sibling-of-parent) where ancestry says so.
3. Radius monotonicity: radius increases with tree depth. Spearman correlation between radius and depth.
4. Manifold validity: every point satisfies ||u|| < 1 in the Poincaré ball, and Lorentz x0 = sqrt(1 + |s|^2) within numerical tolerance.
5. Repeatability: run the benchmark twice, identical input order. Ordering output must match exactly.

Pass conditions (pre-registered, not fitted after):

- Ordering accuracy >= 95%.

- Radius-depth Spearman >= 0.90.
- Manifold validity = 100% of checked points.
- Repeatability = exact match on ordering.

If any condition fails, the geometry-change claim is withdrawn. The model is then, at most, a retrieval / plugin system, not a hyperbolic retraining.

5. Knowledge preservation, as a rate

Definition:

- Take items the base model (A) gets correct on held-out general tasks.
- Preservation rate, per domain: retained correct / base correct.
- Degradation rate: 1 - preservation rate.
- Depth of preservation: preservation rate computed separately by radius tier (center, middle, boundary), then weighted by generality, center weighted highest.
- General tasks and novel tasks are reported separately, never averaged into one hiding number.

Gate for B (Section 6):

- Preservation rate >= 0.98 per domain, aggregated.
- That is, relative degradation <= 2%.
- Example: if A scores 80% on a domain, B must score at least 78.4% (80 x 0.98), not merely "close".
- A domain failing the 2% gate fails B, even if the aggregate passes.

This rate measures preservation, not new learning. New learning is measured separately, in Section 6, and must not be used to hide preservation loss.

6. A / B / C ablation

Arms, identical data, identical order, identical seed, greedy decoding:

Arm	Definition
A	Standard Euclidean model. Glimmer Q4 out of the box. Euclidean attention and retrieval. No map.
B	Euclidean broken, hypo-native retrain, plus knowledge pre-distilled into the hypo map as a plugin. Δ_H trained on H. Map supplies context; the retrained model consumes it.

Arm	Definition
C	Hypo retrained native, plus multimodel map distillation as native / plugin to the agentic harness. One unified map, distilled from multiple models. Map is native to the harness forward: traversal, KV nested in the map, no re-prefill for existing context.

Dropped arm:

- D, multimodel hypo with a hypo-to-Euclidean reader, is DROPPED as an ablation arm.
- Reason: a Poincaré point is already a point in the Euclidean unit ball. Euclidean ranking over it is a reader choice, not a new geometry.
- D collapses to A-style ranking with a conversion step. It adds a conversion, not a distinct mechanism.
- Keep D only as a diagnostic: dual-rank overlap (hyperbolic rank vs. Euclidean rank) reported, not gated as an arm.

Pre-registered gates:

- B vs. A: degradation <= 2% on knowledge preservation, per domain (Section 5). B must also pass G1 geometry acceptance. If B fails either, B is not a hypo-native retrain.
- C vs. A and B: order-of-magnitude gain on the targeted metric, AND preservation still >= 0.98 x A.
  - Targeted metric M: graded reuse rate = fraction of needed context served by traversal without re-prefill.
  - Gate: M_C >= 10 x M_A, and M_C >= 10 x M_B.
  - This 10x gate applies to M only. It does NOT apply to general benchmark accuracy, which is bounded and cannot plausibly rise 10x.
  - If C hits 10x on M but loses more than 2% on general preservation, C fails overall. Targeted gain must not be bought by destroying knowledge.
- Decision routing by preference is logged for all arms, per domain, as a secondary metric. It does not replace the M gate or the preservation gate.

Interpretation rules:

- B within 2% of A, with G1 passed: Euclidean muscle memory broken, knowledge preserved.
- C at 10x M over A and B, with preservation held: native hypo harness beats both plugin and Euclidean forms, on the metric the geometry is actually for.
- Any other outcome is reported as that outcome, not rounded up to the gate.

7. Time necessary for adequate deltas (high-side baseline, not a choice)

Time is gate-driven, not hour-predicted. No Δ_H training throughput has been measured yet, so no hour estimate in this protocol is a measured result.

Adequate delta definition:

- At least one full first-round pass over the domain batch set, each domain at least once.
- G1 and G2 passed.
- Three consecutive checkpoints within tolerance on G1 and G2, not a single lucky checkpoint.
- Receipt logged per checkpoint.

Conservative reserves, erring high. These are mandatory reserves, not targets to beat by cutting training:

Stage	High-side reserve	Notes
Pull + verify MLX Q4	12 hours	17 GB download, retry buffer, index / tensor verification. Preallocated file size does not count as verified.
Extract real representations	12 hours	First batch extraction, placement in H, G1 manifold checks.
First-round Δ_H training	24 hours	Batched safetensors, full forward, checkpointed. Machine runs overnight; this is wall time, not human hours.
A / B / C ablations	12 hours	Identical data and order, greedy, all gates logged.
Routing certification by preference	8 hours	Per-domain routing metrics, final gate rollup.
Total reserve	68 hours	

Rules for this time budget:

- If a stage finishes early, the saved time becomes buffer. It is not cut from the next stage.
- If a stage exceeds its reserve, the certification date slips into the Nov 1-4 buffer. Training time is not compressed to fit Oct 30.
- Oct 30 is a target, not a gate. The gates in Sections 4-6 are gates. The calendar is not a reason to lower a gate.
- Submission deadline remains Nov 4, 9:00 PM ET. Buffer is for packaging, failures, and re-runs, not for starting training.

8. Logging schema (every run)

- Run id, date, base id, delta id, seed, batch order.
- Domain, task set, greedy / sampling flag.
- G1 ordering accuracy, radius-depth Spearman, manifold validity, repeatability.
- Preservation rate per domain and per radius tier.
- M, graded reuse rate, per arm.
- Routing metric by preference, per domain.
- Receipt: source tensors, map version, harness version.

No log, no gate. No gate, no claim.